8x8

EDP 2.0

Reference Data Architecture



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By Smita Debata

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Post End Of July London Meetup !!

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To Start where we concluded ..

Redesigning

Reedesigning involves improving user experience without changing the underlying platform

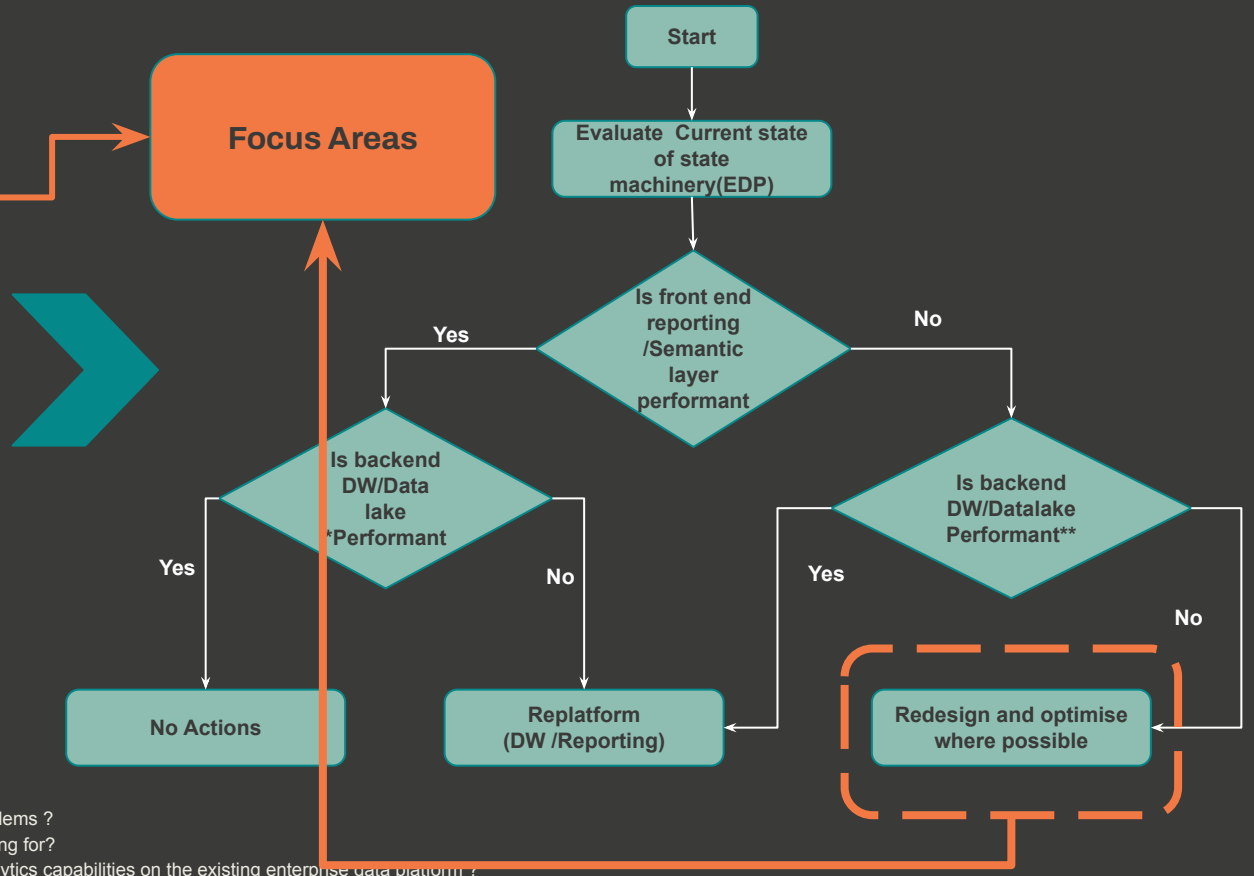
- Focus is on Enhancing self serve experience, better decision and business stakeholder engagement
- Goal is to improve user experience & brand perception
- Consider when current design is inflexible & unable to give right data to right people at the right time

Replatforming

Replatforming involves migrating to a new technology platform

- This focuses on migrating to a new technology platform like new ETL /Database/ hyperscaler etc
- Goal is to improve performance and throughput
- When the current platform is outdated, inflexible, or unable to support business growth

Focus Areas



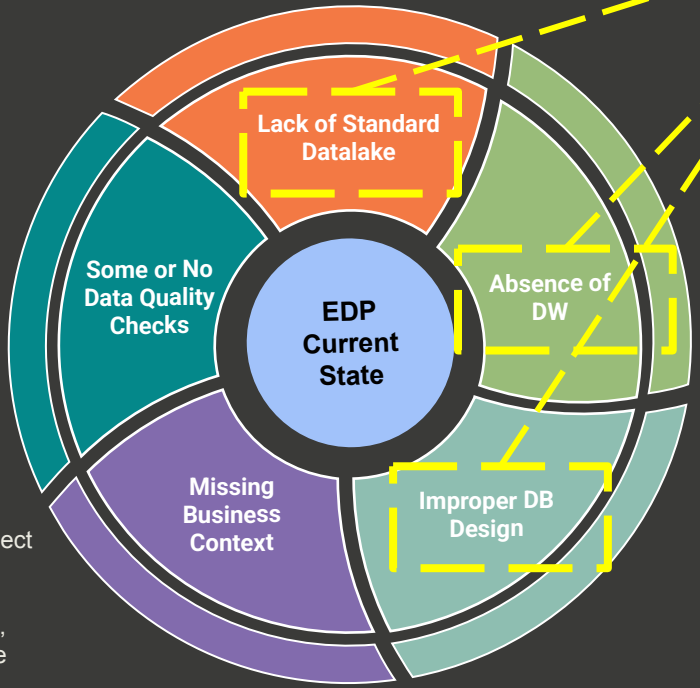
- ★ Is Replatforming alone going to solve our existing problems ?
- ★ Is Replatforming alone going to add value we are looking for?
- ★ Can Replatforming enable AI/ML-driven predictive analytics capabilities on the existing enterprise data platform ?
- ★ Can Replatforming alone enable future vision of empowering our users to “Talk to your Data” using Cortex Analyst (LLM) through Natural language processing (NLP) with the current state of data platform?

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Target Areas from set of Key issues in As - Is EDP

- Not having a standardized data lake architecture (e.g., using Amazon S3 as the central raw data repository).
- Splitted ingestion paths — some directly feeding into Snowflake and others through S3, introduces several **downsides** in terms of loss of data lineage and observability, risk of data duplication and inconsistency
- Reduced reusability(S3-based ingestion allows you to **reuse raw data** for different downstream needs (e.g., analytics, ML, audit)

Priority Areas in redesigning



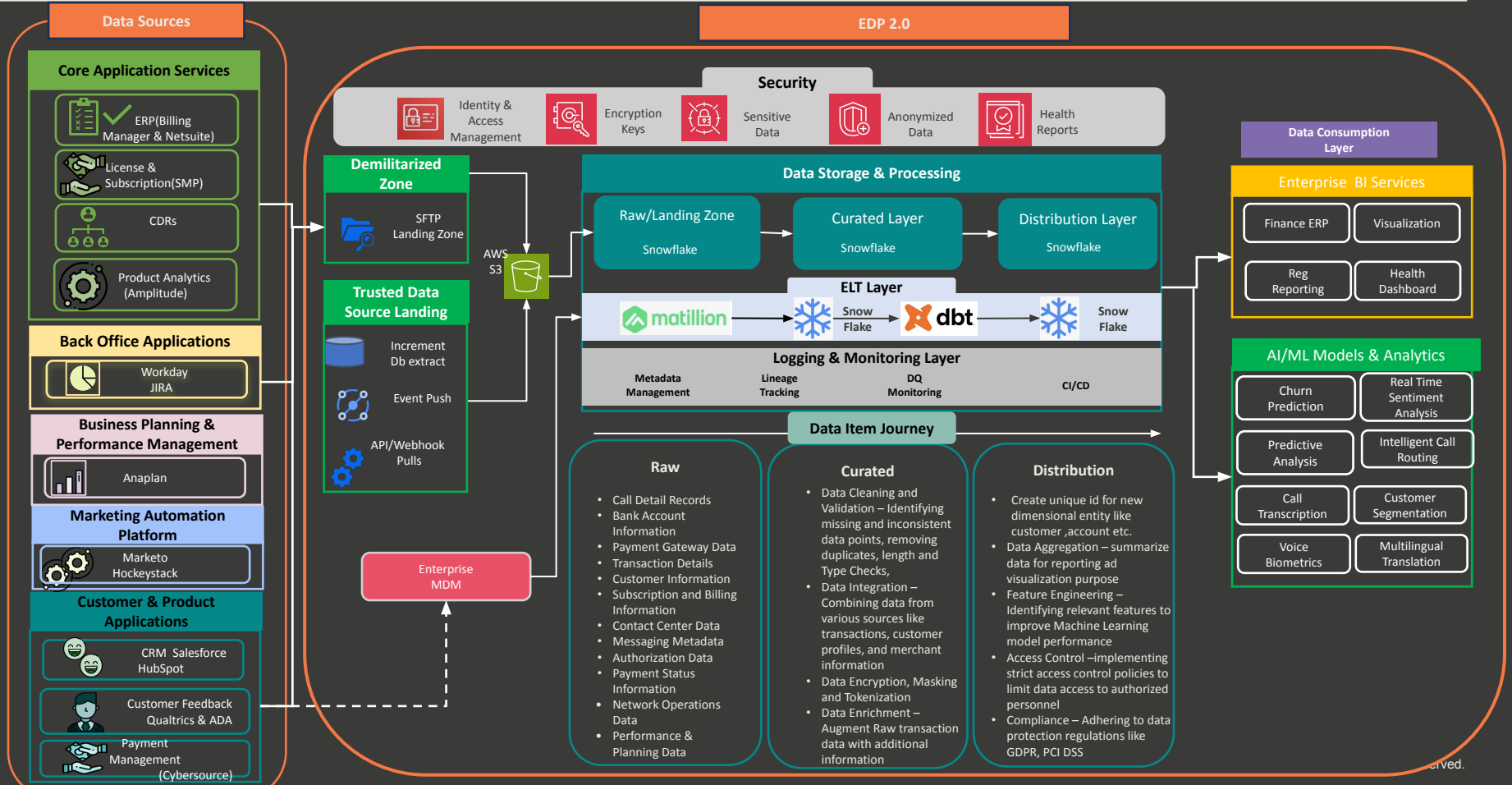
- No Dimension,Fact,Summary,Semantic Layer tables.Hence the need of projects like **Metrics**
- **Under utilising Snowflake** & **Over Utilizing Tableau** with not pride but with disappointment brought from inconsistency in metric values and poor dashboard performances
- No Data Model
- **SCD Implementation** needs to be revisited and been implemented only for a handful of tables for only SFDC source
- Snowflake default time travel is set to only until 1 day
- Using tableau as a DW engine as compared to a semantic engine
- Snowflake default time travel is set to only until 1 day

- No data quality checks during or post ingestion
- **Exception Reporting** should be part of data ingestion and not a project on its own.
- No source to snowflake checks for duplicates , data anomalies, cardinality checks.

- Absence of comprehensive data catalog
- Reverse engineering of EDP produced object level metadata but the business context is missing
- Without business context based dictionary , request for KPI's already existing has to be dealt as new requests

- Creation of Databases /catalog based on projects will keep the chaos adding
- Sticking to Nomenclature based 3-4 layered medallion kind architectures will help support creation of different business objects in a single plane
- performance optimisation best practices missing(like clustering, recommended data types etc)

Reference Data Architecture - High Level



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Next Steps …

- Defining Scope, Planning & Prioritization
- Identifying data entities
- Creating a robust and extensible data model
- Star Schema or a Galaxy Schema or a hybrid of Star & Snowflake
- Identifying Business SMEs to assist in understanding and formulating data model.
- Identifying candidates for redesigning starting from infrastructure,networking needs,aplications

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Q & A

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